DV2657, Report template

Title [A concise, specific, and descriptive title]		
Student 1	**Name**	[Your full name as in LADOK]
	e-Mail	[...@student.bth.se]
Student 2	**Name**	[Your full name as in LADOK]
	e-Mail	[...@student.bth.se]

Abstract
(Approx. 200 words. Write this last. It must be a standalone summary.)

- **Problem:** Briefly state the specific research problem you are addressing.
- **Objective/Motivation:** Explain why this problem is important and what your research aims to achieve.
- **Methodology:** Summarize your high-level approach (e.g., literature review, design of a technical solution, empirical study, data collection and analysis).
- **Expected Outcome/Contribution:** State the anticipated results and their potential impact (e.g., a comparative analysis, a proof-of-concept prototype, a set of validated guidelines).
- **Keywords:** [Keyword 1], [Keyword 2], [Keyword 3], […]

1. Introduction and Motivation

- **Purpose:** To introduce the general field and lead the reader to your specific problem.
- **Content:**
  - Start with the general research area.
  - Provide relevant background facts and define any key terminology.
  - Explain the significance of the area. Why is it relevant? Use statistics or real-world scenarios if possible.
  - Narrow the focus down to the specific sub-area and problem you will investigate.

2. Related Work & Gap Analysis (Mini-SLR)

- **Purpose:** To demonstrate your understanding of existing research and justify your project by identifying a gap.
- **Content:**
  - Synthesize findings from at least 2-3 key research articles. Don't just list them; group them by theme or approach.
  - **Subsection: [Theme 1, e.g., "Existing Techniques for..."]** - Discuss the strengths and weaknesses of these approaches.
  - **Subsection: [Theme 2, e.g., "Evaluation Methods for..."]** - Continue the critical analysis.

- **Gap Identification:** Clearly state the limitations or unanswered question in the current literature that your research will address. (e.g., "While numerous techniques exist, few studies have empirically compared their effectiveness in a real-world context...").

3. Research Questions & Objectives

- **Purpose:** To break down the main problem into answerable, focused questions.
- **Content:**
  - **Main Objective/Aim:** A single sentence stating the overall purpose of your thesis.
  - **Research Questions (RQs):** List 2-4 concrete, answerable questions that will guide your work.
    - **RQ1:** [e.g., What are the most commonly cited XAI techniques in recent literature for increasing user trust?]
    - **RQ2:** [e.g., How do these techniques perform in terms of accuracy, fidelity, and user comprehension when applied to a standard dataset?]
    - **RQ3:** [e.g., What is the measurable impact of the best-performing technique on user trust in a controlled experiment?]

4. Methodology & Research Design

- **Purpose:** To describe *how* you will answer your RQs. This is a core section.
- **Content:**
  - **Overall Approach:** Is your research design experimental, analytical, design-science, case study, etc.?
  - **Data Collection:**
    - Specify your data sources (e.g., public datasets like UCI ML Repository, synthetic data generation, surveys, interviews, code repositories).
    - Justify your choice. How will you ensure data quality?
  - **Technical Solution / Procedure:** Describe the steps you will take.
    - If building something: Describe the system/algorithm design (use a diagram in the appendix if needed), programming languages (Python, Java), and key frameworks/libraries (TensorFlow, Scikit-learn, React).
    - If analyzing: Describe your analytical process (e.g., systematic mapping, comparative analysis, statistical testing).
  - **Evaluation Plan:** How will you measure success? Define your metrics (e.g., accuracy, precision, recall, F1-score, performance latency, user satisfaction scores, thematic analysis codes).
  - **Ethical Considerations:** Briefly discuss any ethical issues (e.g., user data privacy, informed consent, algorithmic bias) and how you will address them.

5. Project Plan & Timeline

- **Purpose:** To show a realistic plan for completing the work.
- **Content:** A table or Gantt chart is highly recommended.
  - **Phase 1: Literature Review & Finalizing Design** (Weeks **)
  - **Phase 2: Data Collection / Implementation** (Weeks **)
  - **Phase 3: Experimentation / Analysis** (Weeks **)
  - **Phase 4: Writing & Drafting** (Weeks **)
  - **Phase 5: Revisions & Final Submission** (Weeks **)

    - Include milestones: "Final RQs defined," "Data collection complete," "First full draft finished."

6. Expected Outcomes & Discussion

- **Purpose:** To state what you expect to find/produce and why it matters.
- **Content:**
    - **Theoretical Outcome:** What new knowledge will be generated? (e.g., "A comparative framework for evaluating XAI techniques.").
    - **Practical Outcome:** What will be built or demonstrated? (e.g., "An open-source implementation of the tested techniques," "A dataset of annotated results.").
    - **Discussion of Impact:** Who will benefit from this work (researchers, practitioners, society)? Discuss the potential **societal and ethical implications** of your findings and the field in general.

7. Limitations

- **Purpose:** To acknowledge the boundaries of your research.
- **Content:** Discuss what your study will *not* cover. This could be related to the scope of RQs, the generality of the dataset used, the simplicity of a prototype, or time constraints.

8. Conclusion & Future Work

- **Purpose:** To summarize the proposal and suggest directions for further research.
- **Content:**
    - Briefly reiterate the problem, method, and expected contribution.
    - Based on your limitations and findings, propose 2-3 logical next steps for future research.

References / Bibliography

- List all cited literature (articles, books, reputable online sources) in a **consistent and standard** citation style (a few standard formats: MLA, APA, Chicago, Harvard, Vancouver, IEEE or ACM).
- This is not just a list; it demonstrates the depth of your literature review.

Appendices (Optional)

- Detailed diagrams or architecture figures.
- Survey or interview questionnaires.
- Preliminary data or results.
- Source code snippets (if highly relevant).

Final submission note: Remove all instructional text, retain the same font and size (Times New Roman, 12), and maintain the page layout. Submit in PDF format, with a total of 8 to 10 pages.